LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Narayan Gurjar
Roll Number	MSE2026002
Programme / Department	MTech IT specialization in software engineering and automation
Topic ID	4___
Full Assigned Topic	Tracking performance drift in Large Language Models Across Successive Version Updates
AI Tool Used	Gemini , Claude
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

● Use only the topic assigned to you by the instructor.
● Generate the paper in a fresh AI conversation.
● Do not tell the AI that you are conducting a citation audit or hallucination test.
● Do not intentionally ask the AI to generate fake references.
● Preserve the original AI-generated paper exactly as produced.
● Do not correct, regenerate, replace, or question references before saving the original output. ● For verification, do not accept another AI chatbot's statement as proof that a reference exists. ● Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources. ● Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references.

You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit

80-110 min Part H: Claim-citation entailment audit 110-120 min Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☑ 2 ☐ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Claude
Model name	sonnet 5
Model/version shown	
Access tier	☑ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☑ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☐ Yes ☑ No ☐ Not sure
Research/Deep Research mode used?	☑ Yes ☐ No
Date/time generated	

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

> Generate a scholarly research paper on the topic: “[YOUR ASSIGNED TOPIC]”. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: ....

Measure	Value
Approximate pages	13
Approximate word count	5886
Number of sections	7

Total number of references 17

Approximate in-text citation occurrences	95-100
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	17
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	11
URL only	0
No persistent identifier supplied	0
TOTAL	28

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI

arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	**Original Bibliography Reference No.**
R1	1
R2	2
R3	3
R4	5
R5	7
R6	8
R7	9
R8	11
R9	12
R10	13

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4 5	Looks genuine Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1		5	yes
R2		5	Yes
R3		5	Yes
R4		5	Yes
R5		5	Yes
R6		5	Yes
R7		5	Yes
R8		5	Yes
R9		5	Yes
R10		5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: "I asked another AI and it said the paper exists." AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publicationexists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y

R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	Y	Y	Y	Y
R8	Y	Y	Y	N	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	AAAI Digital Library / ACM DL / Crossref	DOI: 10.1609/aaai.v33i01.3301 24 29	**Verified accurate.** Bansal, Nushi, Kamar, Weld, Lasecki, & Horvitz (2019), "Updates in Human-AI Teams: Understanding and Addressing the Performance/Compatibility Tradeoff," AAAI Conference on Artificial Intelligence, 33(01), pp. 2429–2437. Authors, title, venue, volume/issue/pages, and DOI all match exactly.
R2	arXiv / Harvard Data Science Review (MIT Press)	arXiv:2307.09009; DOI: 10.48550/arXiv.2307.09 009	**Verified accurate.** Chen, Zaharia, & Zou, "How Is ChatGPT's Behavior Changing over Time?" Confirmed published in Harvard Data Science Review, 6(2) (2024), after arXiv circulation, matching the citation's dual arXiv/HDSR framing. The specific 97.6%→2.4% prime-number statistic quoted in the paper matches the source.

R3	arXiv / ACL Anthology (EMNLP Findings 2024) / Apple ML Research	arXiv:2407.09435; DOI: 10.48550/arXiv.2407.09435	**Verified accurate.** Echterhoff, Faghri, Vemulapalli, Hu, Li, Tuzel, & Pouransari (2024), "MUSCLE: A Model Update Strategy for Compatible LLM Evolution." Authors, title, and arXiv ID all confirmed; also confirmed published inFindingsof ACL: EMNLP2024.
R4	ACM Digital Library	DOI: 10.1145/2523813	**Verified accurate.** Gama, Žliobaitė, Bifet, Pechenizkiy, & Bouchachia (2014), "A Survey on Concept Drift Adaptation," ACM Computing Surveys, 46(4), Article 44. Volume, issue, article number, and DOI all match.
R5	PNAS / PubMed / arXiv	DOI: 10.1073/pnas.1611835114	**Verified accurate.** Kirkpatrick et al. (2017), "Overcoming Catastrophic Forgetting in Neural Networks," PNAS, 114(13), 3521–3526. Full 14-author list, volume/issue/pages, and DOI confirmed.
R6	arXiv	arXiv:2211.09110; DOI: 10.48550/arXiv.2211.09110	**Verified accurate.** Liang, Bommasani, Lee, et al. (2022), "Holistic Evaluation of Language Models" (HELM). Note: this paper was later published in TMLR (2023) rather than remaining an arXiv-only preprint — a minor currency point, but the arXiv citation itself is accurate and the "7 metrics / 16 core scenarios" framing matches the source.

R7	ScienceDirect / Johns Hopkins repository	DOI: 10.1016/S0079-7421(08) 60 536-8	**Verified accurate.** McCloskey & Cohen (1989), "Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem," Psychology of Learning and Motivation, 24, 109–165. Matches exactly (some other citing papers list "1990" or slightly different page ranges, but 1989/109–165 is the correct original).
R8	ACM Digital Library	DOI: 10.1145/3287560.3287596	**Verified accurate.** Mitchell, Wu, Zaldivar, Barnes, Vasserman, Hutchinson, Spitzer, Raji, & Gebru (2019), "Model Cards for Model Reporting," FAT* '19, pp. 220–229. All authors, pages, and DOI confirmed.
R9	arXiv / NeurIPS Proceedings	arXiv:2203.02155; DOI: 10.48550/arXiv.2203.021 55	**Verified accurate.** Ouyang et al. (2022), "Training Language Models to Follow Instructions with Human Feedback" (InstructGPT). Confirmed published in NeurIPS 35, 27730–27744, matching the reference list entry exactly.
R10	arXiv / ACL Anthology (Findings ACL 2023)	arXiv:2212.09251; DOI: 10.48550/arXiv.2212.09 251	**Verified accurate.** Perez et al. (2023), "Discovering Language Model Behaviors with Model-Written Evaluations," Findings of ACL 2023, pp. 13387–13434. Author list (63 authors, abbreviated correctly in the paper), pages, and DOI all confirmed.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A/B/C/D/E	A
R2	A/B/C/D/E	A
R3	A/B/C/D/E	A
R4	A/B/C/D/E	A
R5	A/B/C/D/E	A
R6	A/B/C/D/E	A
R7	A/B/C/D/E	A
R8	A/B/C/D/E	A
R9	A/B/C/D/E	A
R10	A/B/C/D/E	A

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 10 B = 0 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A B	4 3
C	1

D 0
E 1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100
My calculation: 4(10) + 3(0) + 1(0) + 0(0) + 1(0) = 40 points

Authenticity Score = 40 / (4 x 10) x 100 = 100/100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns

60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 100 Total predictions = 100 Prediction Accuracy = 100 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	13
References deeply audited	10
A: Verified	10
B: Wrong metadata	0
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	100 /100
Prediction Accuracy	100 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

No, professional-looking citations do not necessarily turn out to be genuine

2. Did all genuine references actually support the claims for which they were cited?

No, even when a reference is entirely genuine and exists, it does not always support the claim it was cited for.

3. What was the most serious citation-integrity problem you observed?

While the severity can vary based on the specific paper audited, the most serious citation-integrity problem observed in AI-generated academic writing is typically

4. What is the single most important lesson you learned from this lab?

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name:.Narayan Gurjar

Roll Number: MSE2026002

Signature: Narayan Gurjar

Date: 21/08/2026